Activity 1 Requirements to Test Traceability

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Purpose and conclusion

I converted the approved thesis consultation workflow and its main risks into tests with unambiguous expected results. I selected a layered Laravel, Flutter, and live HTTPS strategy because it tests provider behavior, client interpretation, and interoperability. A Postman-only alternative is easier to run but cannot deterministically prove database rollback, widget states, or viewport behavior. The final evidence covers every required normal and adverse category and retains the initial failures that led to hardening.

Assumptions and authority

The test environment is local and nonproduction, all identities are fictional, notification transport is simulated, and Asia/Manila is the evidence timezone while API timestamps remain UTC. The instructor is authorized to accept IA6; the student owns reproduction and explanation. Broader deployment, concurrency thresholds, and external email or SMS are outside this activity.

Traceability summary

Class	Representative test	Expected result	Actual result and evidence
Positive	valid consultation request	201 and one requested record	API-01 passed; workflow test passed
Negative	missing schedule and short agenda	422 with field-safe response	API-04 passed
Boundary	1000-character agenda; per_page=0	maximum accepted once; zero rejected 422	API-05/API-10 passed; APP-02 shows record
Authentication	no bearer token	401 without protected data	API-06 passed
Authorization	student automation; panel receive	403; crypto not attempted for disallowed role	API-07/API-11 passed
Not found	unknown action	404 safe response	API-08 passed
Duplicate	same and changed idempotency key payloads	replay original; changed input 409	API-02/API-03 passed
Timeout	idempotent versus non-idempotent request	same-key retry only when protected	Flutter tests passed
Dependency failure	unused local HTTPS port	bounded error, no loop or data leak	API-13 passed
Recovery	healthy request after unavailable probe	200 and normal data	API-14 passed
UI states	loading, empty, validation, 401, 403, 500, network	visible actionable feedback	22 Flutter tests passed
Responsive	390, 768, 1440 logical widths	no overflow or uncaught error	widget tests passed; APP-04 supports review

The complete 25-row matrix is in `tests/traceability-matrix.md`, and the editable test architecture diagram is in `diagrams/test-architecture.mmd`.

Reproducible method

1. Run the live HTTPS collection with `Invoke-Week15Api.ps1`; it reads the private fixture password but never saves it.
2. Run `Run-Week15Verification.ps1` to execute Laravel, Flutter, analysis, build, audit, and environment capture.
3. Compare `api-results.json`, JUnit XML, and logs with the matrix. Use screenshots only as supporting evidence.
4. Reproduce an initial weakness with the named targeted test, inspect the defect record, and confirm the final full regression.

Interpretation

Observed facts. All 14 live HTTPS cases, 24 Laravel tests with 227 assertions, and 22 Flutter tests passed after hardening. The first targeted run failed five protections, and the first full Flutter rerun exposed a layout regression; both records remain preserved.

Reasoned conclusion. The evidence supports the bounded integration contracts on this working tree and environment. Layered tests found defects that a normal demonstration and a collection-only approach had missed.

Recommendation. Keep the matrix as the change gate, add action-item idempotency, and use Week 16 measurements before selecting rate limits or making any readiness claim.

Limitations and acknowledgment

The evidence does not cover production hosting, public network faults, high concurrency, or all browsers and assistive technologies. AI assistance was used for implementation and documentation and was reviewed against actual source and results. Official Postman and OWASP guidance listed in the module informed the test categories; installed framework behavior was verified locally.